

Deterministic Optimization

Simplex Method in a Nutshell

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How to Walk on the Edge

Simplex Method in a Nutshell

Learning Objective for this module

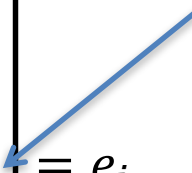
- Compute the direction to move in Simplex method

Adjacent BFS's in Standard Form LP

- Since two adjacent BFS's must share the same set of $n - m - 1$ nonbasic variables as $n - m - 1$ active constraints, and differ in one nonbasic variable
- Example, if $x = (x_1, \dots, x_5)$ has nonbasic variables $x_3 = x_4 = x_5 = 0$, then its adjacent BFS must share two of these three nonbasic variables, i.e. $x_3 = x_4 = x_2 = 0$ may be nonbasic variables in an adjacent BFS
- This means d_N contains 2 zeros and only 1 one component
- In the above example, $d_N = \begin{bmatrix} d_3 \\ d_4 \\ d_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$
- Then $x + \theta d$ will make x_5 positive, i.e. increasing from zero

Figuring Out d_N

- From a BFS x to an adjacent BFS $\bar{x}$, one and only one nonbasic variable of x becomes positive, i.e. becomes a basic variable
- Let's say this nonbasic variable is x_j , then d_N contains all zeros except one at j -th component, i.e.

- $d_N = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \color{red}{1} \\ 0 \\ \vdots \\ 0 \end{bmatrix} = e_j$  j -th component corresponding to x_j

Figuring out d_B

- Once d_N is figured out, the surprising thing is that d_B is uniquely determined
- Recall $Ad = 0$
- So $Bd_B + Nd_N = 0$
- Since $d_N = e_j$ for some nonbasic variable x_j , we have $Ne_j = A_j$, the corresponding column of A
- Therefore, $Bd_B + A_j = 0$
- This gives: $d_B = -B^{-1}A_j$

Which Adjacent BFS to Move to?

- At the current BFS $\bar{x}$, each nonbasic variable x_j provides a direction $d_N = e_j$ and $d_B = -B^{-1}A_j$, pointing to an adjacent BFS
- Which direction should the algorithm pick?
- The algorithm should pick a direction to reduce objective cost
- How does the objective value change along a direction?
 - $c^\top(x + \theta d) - c^\top x$ is the change of objective value
 - $c^\top d$ is the change of objective value for a unit stepsize
 - $\theta c^\top d$ is the total change of objective value after moving θd
- The algorithm should pick a d such that $c^\top d < 0$
- This is called **pivoting**: make x_j enter the basis

Key Concept: Reduced Cost

- $c^\top d$ is very important and deserves a special name
- It's called the **reduced cost** for moving along direction d
- We want to choose the direction d such that $c^\top d < 0$, i.e. reducing objective value by moving along d
- Let's compute $c^\top d$:
 - $c^\top d = [c_B^\top, c_N^\top] \begin{bmatrix} d_B \\ d_N \end{bmatrix} = c_B^\top d_B + c_N^\top d_N$
 - $= -c_B^\top B^{-1} A_j + c_N^\top e_j$
 - $= c_j - c_B^\top B^{-1} A_j$
 - So we denote the reduced cost as $\bar{c}_j = c_j - c_B^\top B^{-1} A_j$.

Key Steps to Implement Simplex Method

- The key step is to find a direction d with two properties:
 1. Pointing from one BFS to one of its adjacent BFS's (compute d)
 2. That adjacent BFS should reduce objective value (compute $\bar{c}_j$'s)
 3. We want to move along the favorable direction as much as possible
 4. Stop when optimal solution is found (or cannot be found)
- We have done 1, 2. Next lecture, we will do 3 and 4.
 - How to move along the direction as much as possible?
 - Need to make stepsize θ as large as possible

Summary

- We learned:
 - How to compute direction to move from one BFS to an adjacent BFS
 - Reduced cost as a criterion to choose a proper direction to reduce the objective value